

Putting Cork Back in *Shoes.*

Can performance footwear be made of a material we usually throw out with the wine?



Each year, Americans buy two billion pairs of shoes and recycle thirteen percent of them. Inside this issue we ask whether a closed-loop bark, harvested every nine years from trees that aren't cut down, can carry the load.

IN THIS ISSUE		CONTRIBUTORS	FIELD NOTES
The Problem at Scale	02	Julian Stoller, Hayden Kim <i>DEA 3030 Materials for Design and Sustainability · Spring 2026</i>	<i>"Cork's been waiting in the wine cellar long enough. Let's put it on the sidewalk."</i>
Material Discovery	03		
Three Designs	04		
A Circular System	05		

Waste at Scale.

The shoe industry sells two billion pairs a year in the United States alone. Most of them end the decade as ash, soil, or stowaway plastic. Here is the breakdown.

Only five percent of American footwear is recycled, which is lower even than the thirteen percent recycling rate for clothing overall. The remainder is incinerated, shipped abroad as second-hand, or buried in landfills where rubber, EVA foam, and bonded textiles can't break down and stay forever.

The U.S. consumes roughly 2.05 billion pairs of shoes per year. At a sober mass estimate of 760 grams each, that's a disposal stream weighing around 3.5 billion kilograms per year. At today's **95 percent landfill-or-incineration rate**, 3.33 billion of those kilograms reach end-of-life without recovery every year.

Per person, the average American buys around eight pairs of shoes a year, which generates roughly ten kilograms of unrecycled footwear waste annually. None of this counts the carbon cost of the petrochemical foams that dominate the modern midsole.

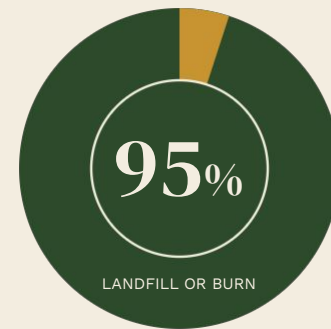
BY THE NUMBERS

3.33B KG OF U.S. FOOTWEAR
LANDFILLED OR
INCINERATED EACH YEAR

2.05B PAIRS OF SHOES
PURCHASED ANNUALLY IN
THE U.S., OR ROUGHLY 6
PAIRS PER PERSON

5% RECYCLED VS. THE 13%
FOR CLOTHING — ABOUT
10 KG PER PERSON OF
WASTE PER YEAR

FIG. 1 END-OF-LIFE SHARE OF U.S. FOOTWEAR, BY MASS



95% LANDFILLED / INCINERATED

5% RECYCLED

PARTIAL SOLUTIONS

TWO OF MANY

Brands have begun displacing petroleum plastic from the upper, but the foam sole has stayed stubbornly the same.



Allbirds

Builds the upper from merino wool and eucalyptus tree fibers, but the high-capacity midsole stays petroleum-derived EVA foam.

The Allbirds product lineup incorporates a variety of sustainable materials but does nothing to address material sustainability in the design of its sole.

Generally Allbirds shoes are highly performant and comfortable as running shoes.



Earthbound

Claims 97% recyclable construction, using materials like hemp, natural latex, cotton, and more.

However this practise of shoe construction limits the brand to low-performance designs, like general walking shoes which also have a limited lifetime.

The problem of footwear waste is vast and in need of more attention. What can we do to address shortcomings in existing solutions? Can we create new footwear designs using a material-first approach? And if we can, will they be competitive as products? Where/how does human-centered design enter the loop?

A Material, Already in the Shop.

Cork is everywhere in footwear — and almost nowhere. We catalog where it lives today, how it's actually bought, and what it can and can't do.

EXISTING IMPLEMENTATION

Cork In Footwear Today

In footwear, cork is broadly limited to men's formal wear, specifically in mens dress shoes, some boat shoes, and women's sandals.



FIG. 2 CATEGORIES WHERE CORK ALREADY APPEARS AT SCALE.

MATERIAL ANALYSIS

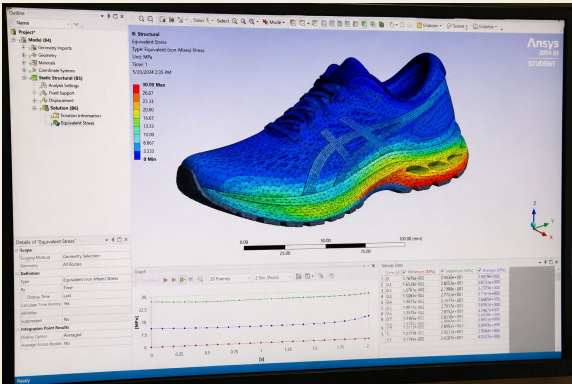
Properties & Caveats

Cork is light, springy, antimicrobial and hydrophobic. So what material properties do we need to design a possible shoe around in order to use cork?

Poor Tensile Strength: Cork is far more susceptible to failure and general degradation when loaded in tension as opposed to compression. Any shoe design that uses cork as a structural material will have to avoid it being loaded in tension.

Low Abrasion Resistance: In pressed cork material, small granules are susceptible to breaking off when exposed to abrasion. Any cork shoe that has cork present on its exterior will have to incorporate features that limit its exposure to the ground.

Low Impact Strength: When struck with force or loaded quickly cork becomes susceptible to fracture. This means any shoe that uses cork as a structurally must not be used athletically.



FORMS OF BULK CORK

How It's Bought, How It's Used

Bulk cork ships in five common forms. Each one is requiring different manufacturing and assembly processes.



Cork sheets come in varying thickness and are often cut into veneers to be used in furniture or sold in thicker sheets to be milled into final shapes. Granules of different sizes (grades) are combined with binders and heat-pressed into various forms.

CORK VS. EVA FOAM

A Side-By-Side Look

EVA is the incumbent midsole. Where cork wins, it wins on circularity and footprint. Where it loses, it loses on stretch and edge wear.

PROPERTY	CORK	EVA FOAM
DENSITY	~0.16 g/cm ³	0.04–0.20 g/cm ³
REBOUND	~90%	55–70%
TENSION	Weak; tears under cyclic load	High; recovers reliably
ABRASION	Lower; granulates wear faster	Higher
ANTIMICROBIAL	Inherent (suberin)	Treated, not native
END-OF-LIFE	Recyclable; bio-derived	Landfill / incinerate

FIG. 3 COMPARISON OF KEY MATERIAL PROPERTIES

Three Shoes, *One Material.*

A generalist runner, a work boot analog, and a heel. Each treats cork differently — as foam, as plank, as accent — and each ships against a benchmark you already own.

The Rocker-Walker

A generalist, all-day comfort shoe.



- PRESSED CORK MIDSOLE
- GRANULATED CORK INSOLE
- RECYCLED MESH UPPER

A set of daily-wear shoes that incorporate cork in such a way that minimizes its degradation with continued use through material placement and the placement of key structures, spearheaded by a generalist daily-wear shoe.

All Wool Upper: Inspired by allbirds wool upper construction we seek to implement this sustainable practices in our flagship shoe.

Natural Latex Outsole: Cork is susceptible to failure upon repeated abrasion. Adding a non-cork outsole allows for the materials use in our assembly without leading to premature degradation.

Target Market: It's a generalist shoe for the general consumer. This shoe is meant for people to walk around in, but not for high performance impact or professional athletic settings.

Limitation: Will likely be harder to manufacture at scale compared to existing shoes of its type. Will also likely not be as springy.

TARGET USER	The walker, the commuter, the all-day-on-feet generalist.
BENCHMARK	Allbirds Tree Runner · On Cloud 5
RISK	High sudden impacts due to highly athletic uses like sprinting.

i. The Boot

A Doc Marten analog — heavier duty, hand-finished.



- PRESSED CORK WEDGE
- CORK SHEET WELT
- VEG-TAN LEATHER UPPER

A pair of shoes intended to mimic the aesthetic and functionality of standard Doc Martens. Little has changed to allow for cork use from a geometrical standpoint, however a structural subsystem has been added to prevent flexure leading to the cork's degradation.

Cork Midsole: Under normal downward loading the cork midsole of a shoe with this design should only be exposed to compressive forces. Like the Rocker-Walker this shoe will also incorporate a latex outsole to prevent material breakdown as a result of abrasion.

Aluminum Shank: To prevent the cork midsole from being exposed to tension while flexuring we embed a recyclable aluminum shank into the midsole. This sub-structure limits the amount that the shoe can bend, minimizing tensile stress.

Target Market: This product is meant to compete with Doc Martens, ideally being worn by people with a loyalty to the aesthetic of Doc Marten shoes or loyalty to the Doc Marten brand should there be cross-business collaboration.

TARGET USER	Trades, women in alternative subcultures, women generally.
BENCHMARK	Dr. Martens 1460 · Red Iron Ranger
RISK	Wet-weather absorption at the welt. Sealed with natural wax.

ii. The Heel

A Demonia heel analog — sculpted column, dressy.



- PRESSED CORK COLUMN
- POWDER-BONDED SHELL
- RECYCLED POLY UPPER

Heels load cork purely in compression, the column never sees lateral stretch. We turn the heel itself out of pressed cork then add a hard but high friction bottom to make the heels more easily walked in.

Blocky Front Sole: This shoe has an all-cork blocky front sole to minimize flexure and subsequent degradation. This also acts as a strong visual statement, incorporating the material used to make the shoe as part of shoe's general aesthetic.

Other Types Of Heels: Despite having a wide variety of designs, women's event-wear heels all share the same property that their ground-facing sections are only loading in compression. This means lots of existing designs can be easily redesigned to use cork. This capacity for easy conversion is particularly visible in women's platform shoes like the Demonia Carmel line and Vermyra Platform Boots.

TARGET USER	Women, high visibility eventwear, occasional fashion piece wearers.
BENCHMARK	Demonia Camel · Steve Madden Slinky · Vermyra Platform
RISK	Tip wear at the stiletto base. Capped with cast brass.

FIG. 4 EACH DESIGN SHOWN ABOVE ISOLATES A DIFFERENT CORK "PLAY" — ANALOG, PLANK, OR SCULPTURE — SO A MANUFACTURER CAN ADOPT ONE WITHOUT RETOOLING FOR THE OTHERS.

A Loop Already Half Built.

Cork moves through one of the cleanest material loops we have. Wine corks are already collected at bars and restaurants by post-consumer networks. We propose footwear plug straight in.

A CORK LIFECYCLE, WITH SHOES IN IT



01 • Harvest

Cork is stripped from the bark of cork oak trees every 9 years. The tree lives on.



02 • Punch & press

Bottle stoppers are punched from the planks; excess bark and trimmings are set aside.



03 • Two lives in parallel

The same stock becomes wine stoppers *and* our shoe components — sole blocks, wedges, heel columns.



04 • Take-back

Used corks and worn shoes return through restaurants, retailers, and a proposed shoe-bin network.



05 • Re-formed

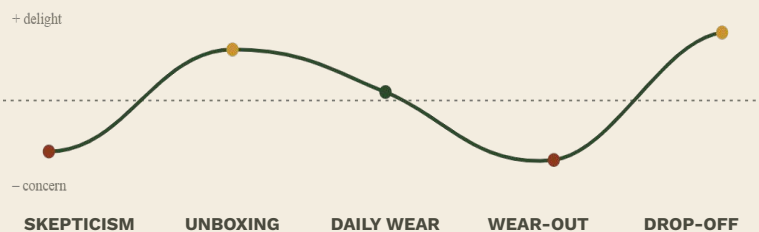
Both streams are ground, re-bonded, and pressed into new shoe soles. Closed.

FIG. 5 MATERIAL FLOW — ADAPTED FROM ESTABLISHED POST-CONSUMER CORK COLLECTION SYSTEMS, WITH A FOOTWEAR-RETURN LEG ADDED.

PROJECTED EMOTIONAL JOURNEY

From Skepticism To Civic Pride.

How we expect a buyer to feel across their first year with the shoe and the two places where take-back matters most.



Consumer starts skeptical of the shoe's performance, but upon unboxing and use they grow to like their shoes before they degrade. However, upon eco-friendly disposal the ecologically-minded user feels solace.

RECYCLING INFRASTRUCTURE

Collaboration With ReCORK:

Cork can be sustainably disposed of through composting, but for the furtherance of the circular economy and sustainability through circular design we can propose specific methods of material reuse.

ReCORK is a Canadian recycling program that engaged in cork collection and material processing (grinding and press-forming) turning cork into granules and eventually insoles.

For our footwear brand, we intend to partner with ReCORK to handle the material processing and for general awareness around cork recycling.

Through this strategic partnership, we can keep cork in the material-use loop for longer, and thereby minimize material waste.

HUMAN-CENTERED INSIGHTS

Three HCD Choices:

Exposed Cork

Visible cork enhances the visuals of our product compared to encased cork. Buyers who care about performance could enjoy the visible material story at the welt or the heel.

Structure Informs Function

By incorporating cork exclusively into shoe designs that have an externally imposed limitation on use (and thereby expected loading conditions) we can guarantee the structural integrity of our material. For instance heels likely won't be used in high performance settings, this means we can use cork in them more viably.

Generalist Focus

All of our design insights exhibit the capacity to be easily reproduced in a wide set of shoes. The properties of heels, Doc Martens, and roller-walkers that make them good for cork substitution are found allow for easy adoption of this material substitution.



JULIAN STOLLER & HAYDEN KIM